

DIGITAL ASSIGNMENT-1

Course Name: Embedded Systems

Course Code: BCSE305L

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Q1)

Compare EDF and RMS with scheduling table and Gantt chart. Give your findings.

Tasks	Execution Time	Time Period
T_1	1	3
T_2	2	4
T_3	1	6

Can you find a schedule for these processes using EDF and RMS scheduling algorithm?

- If yes, draw the Gantt chart and explain.
- If not, explain why?

1) Scheduling using EDF (Earliest Deadline First)

→ Schedulability Test:- $U = \sum_{i=1}^n \frac{C_i}{T_i} \leq 1$

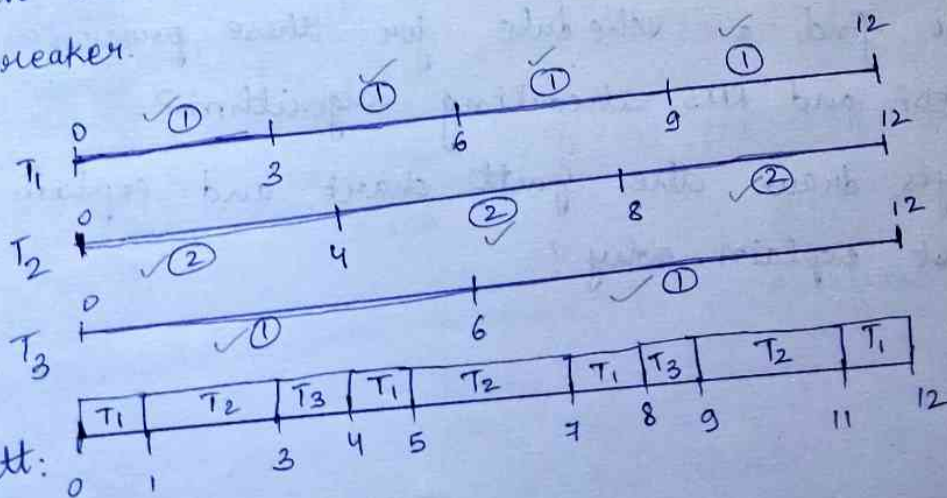
$$\Rightarrow \frac{1}{3} + \frac{2}{4} + \frac{1}{6} = 1 \leq 1 \quad \checkmark$$

Hence, the given tasks can be scheduled using EDF.

→ Calculate Hyperperiod:- $LCM(3, 4, 6) = 12$

→ Process Timelines and Gantt chart:-

Process / Tasks with deadline approaching first will be executed first. Preemption is allowed. In case deadlines are same, we use FCFS as tie breaker.



Gantt:
chart

Explanation:-

- At $t=0$, T_1 executes for 1 ms as the deadline is the closest at 3 ms. [compared to T_2 (4 ms) and T_3 (6 ms)]
- After $t=1$ ms, T_2 is executed as the deadline is at 4 ms, for 2 ms.
- After $t=3$ ms, T_3 is executed till $t=4$ ms as neither T_1 or T_2 arrive.
- For $t=4$ to 5 ms, T_1 is executed since deadline is at 6 ms.
- For $t=5$ to $t=7$ ms, T_2 is executed as deadline is at $t=8$ ms.
- For $t=7$ to 8 ms, T_1 is executed as deadline is at 9 ms.
- For $t=8$ to 9 ms, T_3 is executed as deadline is at $t=12$ ms, all tasks have deadline at 12 ms. But T_3 arrived at $t=6$ ms. [FCFS approach]
- $t=9$ to 11, T_2 is executed since it arrived at $t=8$ ms.
- $t=11$ to 12 ms, T_1 is executed as it arrived last at $t=9$ ms.

∴ Hence, we see that EDF can schedule all the

using

given tasks within their deadlines.

CPU utilization rate :-

$$T_1 = 4 \times 1 = 4$$

$$T_2 = 3 \times 2 = 6$$

$$T_3 = 2 \times 1 = 2$$

hyperperiod i.e.
total time = 12

$$\Rightarrow \frac{4+6+2}{12} = \frac{12}{12} = 100\%$$

(ii) Scheduling using RMS (Rate Monotonic Scheduling)

→ Schedulability test :-

$$U = \sum_{i=1}^n \frac{C_i}{T_i} \leq n(2^{1/n} - 1)$$

$$\Rightarrow \sum \frac{C_i}{T_i} = \frac{1}{3} + \frac{2}{4} + \frac{1}{6} = 1$$

$$n(2^{1/n} - 1) = 3(2^{1/3} - 1) = 0.78$$

$$1 \leq 0.78 \quad \times$$

∴ Hence given set of tasks are not schedulable using RMS as the schedulability test is not satisfied.

→ [∴ Since the combined utilization of tasks is greater than the threshold of tasks]

→ Hyperperiod :- LCM (3, 4, 6) = 12 ms

→ Priority :- $T_1 > T_3 > T_2$

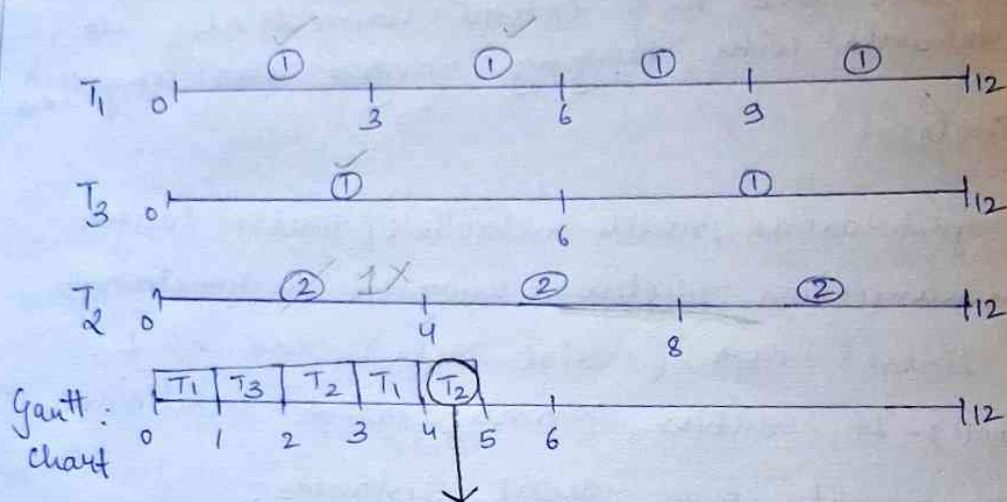
Execution period of $T_1 = T_3$ but Priority $\propto \frac{1}{\text{Time Period}}$

hence, T_1 has higher priority than T_3 .

T_2 has highest execution period, so lowest priority is assigned.

→ Process Timelines and Gantt chart:

Tasks with higher priority are executed first and preemption is allowed.



Explanation:-

→ T_2 has a deadline of $t=4$ ms. But since T_1 which arrives at $t=3$ ms has higher priority than T_2 , it preempts T_2 and executes for 1 ms at $t=3$ to $t=4$ ms.

→ Due to this, T_2 misses its deadline and only executes for 1 ms out of 2 ms execution time. Hence, we can't schedule the given set of tasks using RMS.

→ As T_2 misses its deadline, RMS can't schedule.

Q.2)

Outline the working of I2C and CAN, Embedded Ethernet controller Bluetooth, Zigbee/Wifi.

Sol.

I²C (Inter-Integrated circuit) and CAN (Controller Area Network) are foundational communication protocols in embedded systems, each tailored for specific use cases. Embedded Ethernet controllers, Bluetooth, Zigbee and Wi-fi extend connectivity to broader networks with varying ranges and complexities.

→ I²C Protocol:-

It is a synchronous, multi-controller/multi-target serial communication protocol using 2 bidirectional open drain lines: SDA (Serial Data) and SCL (Serial clock). It enables communication between integrated circuits over short distances.

Data Transmission Process:

- 1) Start condition: The master initiates communication by pulling SDA low before SCL.
- 2) Address Frame: A 7- or 10-bit address is transmitted, followed by a read/write bit (0 for write, 1 for read). Slaves match this address and respond with an ACK (low SDA).

3) Data Frames: 8-bit data packets are sent, each followed by an ACK/NACK bit. The master or slave acknowledges receipt by pulling SDA low.

4) Stop condition: The master halts transmission by pulling SDA high after SCL transitions low-to-high.

Key Features: Supports multiple slaves (upto 128 with 7-bit addressing) using shared bus lines.

- low-speed communication (typically 100-400kHz)
- Requires pull-up resistors on SDA/SCL lines to maintain idle high states.

→ CAN:- CAN is a robust, multi-master protocol designed for automotive and industrial systems. It uses differential signaling (CAN-H and CAN-L) to resist electromagnetic interference.

Working Principle:

- Message Prioritization: Nodes transmit messages with unique identifiers. Lower-ID messages gain priority during arbitration if multiple nodes transmit simultaneously.

Security (WPA)

handles flow control (eg. pause frames) to prevent buffer overflow.

Challenges: • Buffer ...

- Error Handling: Each frame includes a 15-bit CRC. Faulty transmissions trigger automatic retries.

- Frame Types:

- ↳ Data Frames: contains arbitration ID, data (upto 8 bytes), and control bits.

- ↳ Remote Frames: Requests data from another node.

- ↳ Error Frame: Alerts nodes to transmission errors.

Advantages: 1) Fault tolerance via bus redundancy
2) Real time performance with deterministic latency.

→ Embedded Ethernet Controllers: These integrate MAC (Media Access Control) and PHY (Physical layer) modules to enable wired network connectivity in embedded systems.

Data Flow:

1) Transmission: The application generates data, which TCP/IP stack segments into packets.

Buffers in the controller's RAM store packet segments. DMA or PIO transfers data to the PHY for modulation.

2) Reception: PHY demodulates signals, checks

CRC, and stores data in buffers. The controller handles flow control (eg: pause frames) to prevent buffer overflow.

Challenges:

- Buffer Alignment - Non-contiguous data buffers require realignment, often managed via CPU.
- Bus Architecture - Performance depends on reducing wait states during bus transactions.

→ Bluetooth:- Bluetooth uses UHF radio waves (2.4 GHz) for short-range communication between paired devices.

Connection Process:

1. Discovery - Peripheral devices broadcast availability via advertising packets.
2. Pairing - The master (eg. smartphone) authenticates using PINs or cryptographic keys.
3. Data Exchange -

- RFCOMM: Emulates serial ports for simple data streams.
- L2CAP: Manages packet segmentations/reassembly for complex services.

Profiles: Define application-specific roles (eg: A2DP for audio, HID for input devices)

→ Zigbee: Zigbee operates on 2.4 GHz (global), 915 MHz (America's), and 868 MHz (Europe) bands, optimized for low power IoT mesh networks.

Network Architecture:

- Coordinator: Forms the network, stores security keys, and bridges to other networks.

- Routers: Relay data between nodes (mainly -powered)

- End Devices: Battery-operated nodes with minimal functionality

Data Routing:

- Mesh Topology: Self-healing paths ensure reliability. Failed nodes reroute data via neighbors

- AES-128 Encryption: Secures over-the-air

data.

→ WiFi: WiFi uses IEEE 802.11 standards to create wireless LANs via 2.4 GHz, 5 GHz, and 6 GHz bands.

Connection Workflow:

- 1) Beacon Frames: Access Points (APs) broadcast SSID and security protocols.

- 2) Authentication / Association: Devices negotiate

Wifi	
Star	
10+ Gbps	
Low	High
Next	Br
Home	A

Security (WPA 2/WPA 3) and exchange IP addresses via DHCP.

3) CSMA/CA: Nodes avoid collisions by waiting for idle channels before transmitting.

Performance Factors:

- Frequency vs Range - 2.4 GHz offers longer range; 5/6 GHz provide higher throughput.
- Modulation - OFDM (Orthogonal Frequency-Division Multiplexing) splits data into subcarriers for efficient transmission.

Comparative Overview: Each protocol addresses distinct needs, from FC's simplicity in chip communication to Wi-Fi's high-speed internet access, reflecting a trade-off between power, range, and data throughput.

Feature	I ² C	CAN	Zigbee	Wi-Fi
Topology	Bus	Bus	Mesh	Star
Speed	≤ 3.4 Mbps	≤ 1 Mbps	20-250 kbps	10+ Gbps
Power Use	Low	Moderate	Very low	High
Use Case	IC communication	Automotive systems	Smart Home	Internet Access

Q3) Discuss the Challenges & constraints of special purpose computing system with a case study.

Sol.ⁿ Special purpose computing systems are designed for a specific function rather than general purpose computing. These systems include embedded system, industrial control systems, and high-performance computing for specialized applications such as medical imaging, automotive control or aerospace navigation. While they offer efficiency and optimization for their intended purpose, they also come with several challenges and constraints:

Challenges & Constraints:

1. Design Complexity: special purpose systems require highly optimized hardware and software, making the design phase complex. Custom architectures may need to be developed from scratch, requiring significant expertise.
2. Limited Flexibility and Upgradability: Unlike general purpose systems, these systems cannot be easily reconfigured or upgraded.

Hardware and software modifications may

require significant redesign efforts.

3. High Development cost: Custom Hardware and Software increase development costs.

low production volume can further increase per-unit cost compared to mass produced general purpose systems.

4. Power & Performance Constraints: Many special purpose systems, such as embedded and IoT devices, operate under strict power constraints.

Trade off between computational performance and energy efficiency must be carefully managed.

5. Real Time Constraints: Some applications, such as automotive control or medical devices, require real time responses with minimal latency. Delays in computation can lead to catastrophic failures in critical systems.

6. Security and Reliability Issues: Security vulnerabilities in specialized systems can be harder to patch, as updates may not be frequent.

Reliability is crucial, especially in mission-critical applications such as aerospace and healthcare.

7. Compatibility & Integration Challenges: These systems often need to integrate with other computing environments, which may lead to compatibility issues.

Standardization is limited, making interoperability between different special-purpose systems difficult.

Case Study:- Real Time Embedded system in Automotive Control

Background: A modern vehicle's Engine control unit (ECU) is a special-purpose computing system that manages engine performance, fuel injection, emissions control, and more. The ECU must process sensor data in real time to ensure the vehicle runs efficiently and safely.

Challenges Faced:

1. Real time processing-

- ECU must react within milliseconds to changes in engine parameters to optimize fuel efficiency and emissions.

- Any delay could result in poor performance, increased fuel consumption, or even engine damage.

2. Hardware & Software optimization:

- The ECU uses specialized mc & software tailored to a specific vehicle model.
- updating the software for different models or regulations requires significant testing and validation.

3. Power Constraints:

- The ECU operates within a limited power budget to avoid excessive battery drain.
- optimization techniques, such as low-power sleep modes, must be incorporated.

4. Security Risks:

- Modern vehicles with internet connectivity are vulnerable to cyberattacks.
- Ensuring secure communication b/w the ECU and other vehicle systems is critical.

5. Durability & Reliability:

- The ECU must function under extreme environmental conditions (heat, vibration, moisture)
- Failure can lead to significant safety risks.

Solutions Implemented:

- RTOS: used to meet strict timing requirements
- Efficient Power Management: low power processors and sleep modes optimize energy usage.
- Robust Security Measures: Encrypted communication and intrusion detection systems enhance cybersecurity.
- Thorough testing & Validation: Extensive simulations and real-world testing ensure reliability.

Conclusion: Addressing these challenges require

careful consideration of hardware, software and system integration to ensure efficiency, reliability, and security.

9
Outline the current trends in Embedded systems, their future directions, applications and advancements.

Current trends in Embedded Systems:-

→ Integration of AI/ML:

ES are becoming equipped with AI and ML algorithms that allow real-time, on-the-edge decision-making with data processing; for example, in smart cameras with object detection, low power μ c are getting optimized to run lightweight ML models.

Eg:- AI powered Health monitoring wearables.

→ Proliferation of IoT:

ES are the very backbone of IoT world. They allow connectivity and sharing of information in smart homes, cities, etc. Protocols like MQTT, CoAP, Zigbee allow efficient exchange of information. Eg:- Well known and common applications are smart thermostats and industry sensors.

→ Energy Efficiency and Reduced Consumption:-

Ultra low power μ c enhance survivability by using energy harvesting techniques such as solar energy. Dynamic voltage scaling and sleep modes are also common. Eg:- Green batteries, battery powered environmental sensors.

→ Miniaturization & System Development on Chip:-

Semiconductor tech advancements achieving smaller, powerful ES with embedded CPU, memory, peripherals on one chip. eg: Raspberry Pi Pico, ESP32 SoCs.

→ Real-Time Processing Refers to Edge Computing:-

Most of online processing are possible when applications use data to be processed closer to the place where they are produced rather than cloud. Eg:- RTOS-like FreeRTOS, Zephyr-based systems.

Future Direction of ES:-

→ Pervasive AI at the edge: Future ES will have advanced AI capabilities that make it possible for systems to work autonomously.

Eg: Development of neuromorphic chips

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→ Future connectivity of 5G and beyond: ultra low latency, high bandwidth communication in embedded devices when integrated with 5G & 6G networks.
Improved performance on autonomous vehicles & telemedicine.

→ Quantum Computing Influence: Quantum inspired algo. may eventually filter down into ES for estoric applications such as cryptography.
Hybrid systems that combine classical & quantum processing.

→ Sustainability & Green ES: Materials will be biodegradable or recyclable; hardware designs will generate less e-waste. Self powering systems will tap ambient energy sources & dominate the future IoT landscape.

→ Human Machine Integration: ES will feature prominently in bioelectronics, such as brain-machine interfaces & advanced prosthetics. Seamless integration of human & machine. for greater enhancement in medicine.

→ Customizable & Modular Designs: Transitioning towards open source hardware & software platforms like RISC-V because it offers flexibility & cost savings. Rapid prototyping & deployment through modular embedded platforms.

Applications of ES:-

- Consumer Electronic Devices: Smartphones, Smart TVs and wearables. Acting: Process User inputs, manage displays & enable communication.
- Automotive System Devices: ECUs, ADAS. Acting: RT monitoring, autonomous driving, V2V communication.
- Healthcare Devices: Pacemakers, glucose monitors, medical imaging systems.
- Industrial automotive Devices - PLC, robotic arms.
- Aerospace & Defense Devices - Flight control system, radar sys, navigation surveillance.
- Smart Infrastructure Device: Traffic Management sys, smart grids.

Developments in ES :- ↳ Sensor Technology.

Miniaturized, high precision sensors (LIDAR). Eg: gesture recognition in gaming controllers.

↳ Wireless communication Improvements: Adoption of Wifi 6, Bluetooth 5.2 for long range & reliable comm. Eg: LoRa-based agricultural sensors.

↳ Software Ecosystem Growth: Robust RTOS & middleware. Eg: Zephyr OS.

↳ Improvements in processors & hardware: development of multi core processors & GPUs for HLL processing. Eg:

NVIDIA Jetson Nano for AI-enabled embedded applications

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- ↳ Power Management Innovations:- changes in battery technology (eg: solid state batteries) & power efficient architectures. Eg:- wearable with battery life lasting weeks.
- ↳ Security Upgrades:- hardware encryption, secure enclaves, and over the air patches for security vulnerabilities. Eg:- Secure IoT gateways in smart homes.

Q.5) Discuss one case study application of Embedded System from each domain in detail (Agriculture sector, Automotive / consumer / Industrial controls / Medical Electronics)

Sol. 1) Agriculture sector: Smart Irrigation System
Case Study: Automated Soil Moisture-Based Irrigation System.

↳ Problem: Traditional irrigation systems waste water by irrigating fields regardless of soil moisture levels.

Embeddel System Application: An embedded MC-based system automatically monitors soil moisture levels and controls the irrigation pump accordingly.

Components:

- μc : Arduino or STM32
- Soil moisture sensor
- Water Pump & Relay Driver
- Solar Panel (for rural deployment)
- GSM module (for remote alerts)

Working:

- The μc continuously reads data from soil moisture sensor.
- If moisture drops below a threshold, the μc triggers the water pump.
- After adequate moisture is reached, pump is turned off.
- Farmers receive SMS updates on irrigation status via GSM.

Benefits:

- 1) Conserve water
- 2) Reduce manual labor
- 3) Increase crop yield & sustainability

2. Automotive Sector: Airbag Control System

Case Study: Embedded Airbag Deployment Controller

Problem: During high-impact collisions, lack of timely airbag deployment leads to severe injuries or fatalities.

Embedded system Application: A real time ES monitors crash sensors and deploys airbags within ms of detecting a collision.

Components:

- 1) μC
- 2) Accelerometers & Gyroscopes
- 3) Crash Sensors
- 4) Airbag Inflator circuit

Working:

- Sensors detect rapid acceleration
- The embedded ECU processes this data in real time.
- If crash parameters exceed thresholds, ECU sends a trigger to deploy airbags.
- Entire process occurs within 20-50 ms.

Benefits:

- 1) Saves lives by minimizing collision impact
- 2) Rapid, reliable real-time response
- 3) Self-testing ensures system readiness at all times.

3. Consumer Electronics : Smart Washing Machine

Case Study: Fuzzy logic-Based washing machine controller.

Problem: Conventional washing machines operate on pre-defined cycles regardless of load type, leading to inefficient washing & energy waste.

Embedded system Application: A smart washing machine uses ES with fuzzy logic to optimize wash cycles based on fabric type, load weight, dirt level.

Components:

- MC (8051, ARM Cortex-M)
- Weight sensor, Water level sensor
- Turbidity sensor
- Motor Driver & Display Unit

Working:

- Sensors detect the weight, water clarity, and type of fabric
- the controller selects appropriate water levels, detergent quantity, and wash time
- users get real-time status updates on display.

Benefits:

- 1) saves water, power, time.
- 2) Enhances user experience with automation
- 3) Improves wash quality.

4. Industrial Controls : PLC-based Conveyor Belt Sorting System.

Case Study: Color-based Object sorting in Packaging Industry

Problem: Manual sorting of objects based on color or type is time consuming and error prone.

Embedded System Application: A PLC controls a conveyor belt system to automatically sort items based on color.

Components:

- PLC (eg Siemens S7-1200)
- Color sensors/Camera
- Actuator (Pneumatic Pusher)
- Conveyor Belt Motors

Working:

- Items move along conveyor belt.
- A sensor/camera captures object color.
- Based on detected color, PLC activates specific actuators to divert objects to appropriate bins.
- Entire process is real time and continuous

Benefits:

- 1) Increases speed & accuracy of sorting.
- 2) Reduces labour cost
- 3) Easily scalable & programmable for various items.

5. Medical Electronics : Wearable Heart Rate Monitor

Case Study: Continuous Heart Rate Monitoring using ES.

Problem: Patients with heart conditions require continuous heart monitoring, especially outside clinical environments.

Embedded System Application: A wearable embedded device measures heart rate & alerts patients/ doctors in case of abnormalities.

Components:

- μC (ARM Cortex-M)
- Pulse Sensor / ECG Module
- Bluetooth / Wifi / GSM module
- Batteries + OLED display

Working:

→ Sensor continuously monitors pulse rate and transmits data to a mobile app.

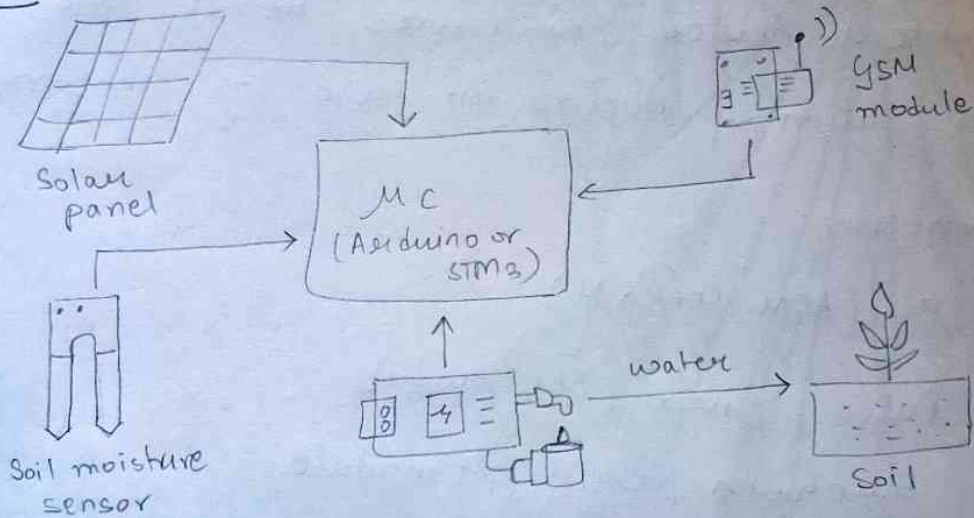
→ If abnormal heart activity is detected (eg:- arrhythmia) alert is sent to doctor or emergency contact.

→ Data is stored on cloud for remote monitoring.

Benefits:

- 1) Enables proactive healthcare
- 2) Reduces hospital visits.
- 3) Improves quality of life for patients

Diagrams:



Smart Irrigation System

Airbag control System :

